



PROJECT ABSTRACT

Campus Issue Reporting and Resolution System

Submitted in partial fulfilment of the requirements of the Adaptive Software Engineering (ASE) course

Project Title	Campus Issue Reporting and Resolution System (CIRRS)
Course	Adaptive Software Engineering (ASE)
Guide Name	Ms. Bhavya
Development Approach	Agile / Adaptive, Incremental Sprint-based Delivery

Team Members

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1. Introduction

Educational campuses generate a continuous stream of day-to-day operational issues — malfunctioning Wi-Fi routers, faulty classroom electronics, plumbing failures in hostels, unclean restrooms, broken furniture, and irregular transport schedules. At present, most institutions rely on informal channels such as verbal complaints, phone calls, or physical registers maintained by administrative staff to report and track such issues. This process is slow, undocumented, and lacks accountability, which frequently results in delayed resolution, duplicated complaints, and dissatisfied students and staff. The Campus Issue Reporting and Resolution System (CIRRS) is proposed as a centralized, web-based platform that allows students and staff to report campus-related issues digitally, track their resolution status in real time, and enables administrators to prioritize, assign, and resolve issues efficiently.

2. Problem Statement

There is currently no unified, transparent, and trackable mechanism through which students can report campus infrastructure or facility-related issues and receive timely updates on their resolution. Complaints raised informally are often lost, forgotten, or unaddressed due to the absence of a structured logging and escalation system, leading to recurring inefficiencies and a poor overall campus experience.

3. Objectives

- To provide a simple, accessible interface for students and staff to report campus issues along with category, description, location, and optional photo evidence.
- To maintain a transparent status pipeline (Reported → Acknowledged → In Progress → Resolved) that is visible to the person who raised the issue.
- To provide administrators with a centralized dashboard to view, filter, prioritize, and assign incoming issues to the relevant department.
- To reduce average issue-resolution time through better visibility, accountability, and structured escalation.
- To design and build the system using an adaptive, iterative development methodology that accommodates evolving requirements across sprints.

4. Proposed System

CIRRS will be developed as a role-based web application with two primary interfaces. The student/staff-facing interface allows any campus member to log in, submit a new issue report with category and location tags, attach a photograph if relevant, and track the live status of previously submitted reports. The administrator-facing interface presents a consolidated dashboard of all open and resolved issues, with filtering by category, urgency, and department, along with the ability to update status, add resolution notes, and reassign issues where necessary. Automated status-change notifications keep the reporting user informed without requiring them to follow up manually.

5. Adaptive Software Engineering Approach

In line with the Adaptive Software Engineering methodology, CIRRS is developed through short, iterative sprint cycles rather than a fixed upfront specification. Each sprint delivers a working increment of the system, which is reviewed and adjusted based on feedback before the next cycle begins. This approach is well suited to the project because real requirements naturally emerge and evolve during development — for example:

- Sprint 1 (Core Reporting): Basic issue submission form and a persistent database to store reports.
- Sprint 2 (Status Tracking): Status pipeline and a personal dashboard for users to track their own submissions.
- Sprint 3 (Admin Console): Centralized admin dashboard with filtering, assignment, and resolution notes.
- Sprint 4 (Adaptive Enhancement): Requirements evolved based on user feedback to add photo-evidence uploads and category-wise priority handling — demonstrating the system's adaptability to changing needs rather than a rigid, pre-fixed scope.

This incremental delivery model, backed by a version-controlled Git repository with sprint-wise commit history, allows the design to adapt continuously to real feedback while maintaining a working, demonstrable product at every stage.

6. Technology Stack

- Frontend: HTML5, CSS3, JavaScript (React)
- Backend / Database: Firebase (Authentication, Firestore Database, Hosting)
- Version Control: Git and GitHub for iterative, sprint-wise source control
- Design: Responsive, mobile-first UI accessible from any campus device

7. Expected Outcomes

- A fully functional, centralized issue-reporting platform accessible to all campus users.
- Reduced average resolution time due to transparent tracking and administrative accountability.
- A documented, sprint-wise development history demonstrating practical application of adaptive software engineering principles.
- A reusable, scalable foundation that can be extended to additional departments or institutions.

8. Conclusion

The Campus Issue Reporting and Resolution System addresses a genuine, everyday operational gap within educational institutions by replacing informal, untracked complaint channels with a transparent, centralized digital workflow. Beyond its practical utility, the project serves as a direct application of Adaptive Software Engineering principles, with its sprint-based, feedback-driven development process reflected in the evolving feature set and the version-controlled commit history of the project repository.